

Project Practice for Deep Learning Project1: Exploring Supervised and Foundation Models for Image Segmentation

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Instructions:

1. **Deadline:** 2025-10-19 11:59 pm
 2. You need to write LaTeX. Word is ok, but PDF! Give your solution in English.
 3. Your report should be submitted in PDF format and packed with your code, and the naming format of the file is `studentID-name-project1.zip`.
 4. Please submit your zip file to <https://epan.shanghaitech.edu.cn/1/JFrYpF>
 5. **Carvana dataset can be found here:** <https://epan.shanghaitech.edu.cn/1/F2JPuU>
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1 Introduction

This project focuses on exploring different paradigms of image segmentation, ranging from traditional supervised learning to state-of-the-art foundation models. In the basic task, we train and evaluate a U-Net model [4] on the Carvana dataset [1], a well-established benchmark for binary semantic segmentation. This step provides hands-on experience with data loading, hyperparameter setting, model training, evaluation metrics, and visualization, highlighting both the strengths and limitations of supervised methods in handling a specific, well-defined problem.

Building upon this foundation, the advanced tasks shift towards the exploration of foundation models. We first analyze the zero-shot segmentation capability of the Segment Anything Model (SAM) [5], examining its success cases and limitations when applied to real-world images. Subsequently, we extend SAM with GroundingDINO [2] to achieve text-driven semantic segmentation, demonstrating how a class-agnostic model can be upgraded into a semantically aware one through multimodal integration. Collectively, these tasks provide a comprehensive perspective on the evolution of segmentation techniques, from specialized supervised networks to general-purpose foundation models.

2 Basic Task

Your primary goal in this task is to train and evaluate a Pytorch U-Net[3] model for segmenting cars from the background in the Carvana dataset [1]. To begin, you will need to clone the official Pytorch-UNet repository [3] and place the downloaded Carvana dataset into the appropriate project path.

A critical point to note is that **running the training script with the repository's default settings may cause the loss to become NaN (Not a Number)**. Therefore, your first practical step is not just to

run the code, but to inspect the training script and adjust key hyperparameters, such as the `learning rate` and `batch size`, to ensure a stable training process. Once you achieve stable training, your objective is to run inference on the test set (you can collect some car photos by yourself as a test set). In your report, be sure to present your final Dice score and a few visual examples comparing the input image and your model's prediction. **If everything goes well, the validation Dice should be over 0.90.**

This task will give you a deep appreciation for the strengths (high accuracy) and weaknesses of traditional supervised learning methods. Critically, the trained model can only be used for one specific task (segmenting cars). This limitation is precisely the problem that Foundation Models, which we will explore in the advanced tasks, aim to solve.

3 Advanced Tasks

Exploring the Zero-Shot Power and Limitations of SAM. Create your own mini-dataset by taking 20 photos. Use SAM's "everything" mode [5] to perform automatic segmentation. Your goal is to identify and summarize its failure cases. Additionally, analyze its `multimask_output` mode, observing how it produces masks at different scales or levels of detail. This will provide a first-hand understanding of SAM's powerful zero-shot capabilities and teach you how to pinpoint the shortcomings of a State-of-the-Art (SOTA) model in complex, real-world scenarios (e.g., with complex textures, transparent objects, or small objects).

From Class-Agnostic to Semantic Segmentation with GroundingDINO + SAM. Build a GroundingDINO + SAM pipeline [2, 5] to obtain precise segmentation masks for objects based on a text prompt. This task upgrades a class-agnostic segmenter (SAM) into a semantic segmentation model that understands natural language. You will learn how to use the output of one model (a bounding box) as a prompt to efficiently drive another model.

References

- [1] *Carvana Image Masking Challenge*. <https://www.kaggle.com/competitions/carvana-image-masking-challenge>. Accessed on September 23, 2025.
- [2] IDEA-Research. *Grounding DINO*. <https://github.com/IDEA-Research/GroundingDINO>. Accessed on September 23, 2025.
- [3] Milesial. *Pytorch-UNet*. <https://github.com/milesial/Pytorch-UNet>. Accessed on September 23, 2025.
- [4] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. "U-Net: Convolutional Networks for Biomedical Image Segmentation". In: *arXiv preprint arXiv:1505.04597* (2015).
- [5] *Segment Anything Model (SAM)*. <https://segment-anything.com/>. Accessed on September 23, 2025.